Design and DC Analysis of a GAA Si Nanowire nFET

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***Abstract*—An n-type gate-all-around (GAA) silicon nanowire transistor with a 6 nm diameter and a 15 nm gate is simulated in Synopsys Sentaurus using a two-dimensional r–z half-slice under a cylindrical coordinate system, so that terminal currents are obtained in amperes for a complete wire rather than per unit width. The gate stack is a 0.5 nm SiO_2 interfacial layer with 2.256 nm of HfO_2, giving a conventional equivalent oxide thickness (EOT) of 0.9 nm. Source/drain doping is engineered as complementary error-function tails from the inner-spacer edges, placing the metallurgical junctions 0.24 nm inside the gate edges and giving a 14.51 nm effective channel. Drift-diffusion transport with density-gradient quantum corrections, band-to-band tunnelling, thin-layer and remote-phonon mobility degradation, and 1 GPa uniaxial tensile stress yields $I_{ON} = 1.459 \times 10^{-5}$ A, $I_{OFF} = 1.936 \times 10^{-11}$ A, peak $g_m = 4.137 \times 10^{-5}$ S, and a maximum electron velocity of 2.078×10^7 cm/s at $V_D = V_G = V_{DD} = 0.7$ V.**

***Index Terms*—Gate-all-around, nanowire FET, cylindrical coordinates, equivalent oxide thickness, band-to-band tunnelling, density gradient, TCAD.**

I. Device Structure and Simulation Setup

The device is an n-channel GAA nanowire, built entirely by process simulation through 2-D region tiling of the r–z half cross-section. Structural and material parameters are collected in Table I. Fig. 1(a) shows the simulated half-slice: the radius runs horizontally from the symmetry axis at the left edge and transport runs vertically, so revolving the slice about that edge recovers the complete wire. Visible are the n^+ source and drain epi, the undoped channel, the SiO_2 interfacial layer and HfO_2 shell, the low-k inner spacers, and the three contacts.

A. Meshing

A base grid is laid down by tagged coordinate lines at every region boundary, with the spacing graded from 0.5 nm in the source/drain epi to 0.15 nm across the silicon surface. On top of this, a single adaptive refinement box is applied *only to the channel and oxide material*, targeting element edges of 0.2 nm axially and 0.15 nm radially and subdividing wherever the net-doping gradient exceeds a fixed threshold. The dashed boxes in Fig. 1(b) and (c) mark its extent.

Two considerations drive this. The inversion charge is confined to a shell a few tenths of a nanometre thick at the silicon surface, so the radial spacing must resolve it. More importantly, band-to-band generation is exponential in the local electric field and I_{OFF} is a scored metric: a coarse mesh at the reverse-biased drain junction under-resolves the field peak and silently under-reports leakage.

TABLE I
Device structural and material parameters

Parameter	Value
Gate metal length (L_G)	15 nm
Wire Radius (R)	3 nm
Inner spacer (L_{sp})	4 nm
Source/drain epi length (L_{SD})	10 nm
SiO_2 interfacial layer (t_{IL})	0.5 nm
HfO_2 thickness (t_{HK})	2.256 nm
Equivalent oxide thickness (EOT)	0.9 nm
Gate metal workfunction (Φ_M)	4.4 eV
Channel doping (N_{ch}, UID)	10^{15} cm^{-3}
Source/drain doping (N_{SD})	4×10^{20} cm^{-3}
Doping decay length (L_{diff})	1.27 nm
Effective channel length (L_{eff})	14.51 nm
Uniaxial stress, transport direction	1 GPa (tensile)
Supply voltage (V_{DD})	0.7 V
Temperature (T)	300 K
Spacer permittivity (ϵ_{LowK})	5.0
HfO_2 permittivity (ϵ_{HK})	22.0

B. Why a 2-D Slice Is a 3-D Wire

The device simulator is invoked with

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Math{ CoordinateSystem{DFISE}  Cylindrical(0) }
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This does not create a solid of revolution. It weights every control volume by $2\pi r$, its distance from the rotation axis, following Pappus's theorem. Two consequences matter for the reported metrics.

First, terminal currents emerge in **amperes for one complete wire**, not in A/μm, which is exactly the unit required for I_{ON} and I_{OFF}; the area factor is left at unity. A Cartesian 2-D run of the same slice would instead report $I_{ON}/(2\pi r_{\text{inv}}) \approx 9.3 \times 10^{-4}$ A/μm, taking the inversion radius as $r_{\text{inv}} \approx 2.5$ nm.

Second, the symmetry axis at $r = 0$ requires **no boundary condition**. Elements touching the axis carry vanishing control volume under the $2\pi r$ weighting, so reflective symmetry is enforced by the discretisation itself. This is why the rotation axis x_0 must satisfy $x_0 \leq \min(x)$ over the device; here $x_0 = 0$.

II. Calculation of the Equivalent Oxide Thickness

A. Conventional (Planar) Definition

Treating the stack as series parallel-plate capacitors of equal area, the inverse capacitance per unit area is

$$\frac{1}{C} = \frac{t_{IL}}{\epsilon_0 k_{IL}} + \frac{t_{HK}}{\epsilon_0 k_{HK}}. \tag{1}$$

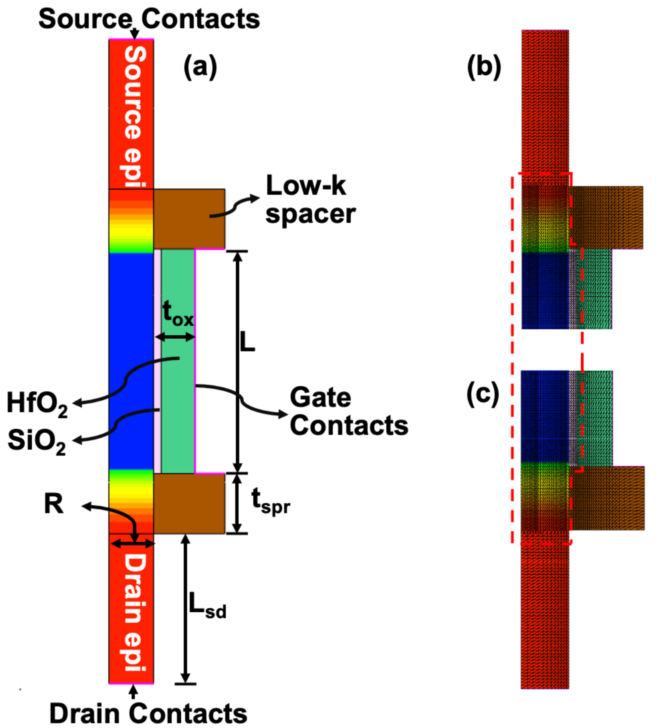

Fig. 1. (a) Material regions with doping overlaid. (b), (c) Mesh at the source- and drain-side spacers; dashed boxes mark the adaptive refinement.

Equating this to a pure SiO_2 layer of thickness EOT gives the standard expression, which is what we build to:

$$\mathrm{EOT} = t_{IL} + t_{HK}\,\frac{k_{ox}}{k_{HK}} \implies t_{HK} = \left(\mathrm{EOT} - t_{IL}\right)\frac{k_{HK}}{k_{ox}}. \tag{2}$$

With EOT $= 0.9$ nm, $t_{IL} = 0.5$ nm, $k_{ox} = k_{IL} = 3.9$ and $k_{HK} = 22$,

$$t_{HK} = (0.9 - 0.5) \times \frac{22}{3.9} = 2.2564 \text{ nm}. \tag{3}$$

The resulting radii are $R_1 = 3.0000$, $R_2 = 3.5000$, $R_3 = 5.7564$ and $R_4 = 7.7564$ nm for the silicon surface, the IL/HfO_2 interface, the HfO_2/metal interface, and the metal outer edge. The value $k_{HK} = 22$ is asserted identically in the process deck and in the `sdevice` parameter file; a mismatch would leave the built geometry and the reported EOT silently inconsistent.

B. Coaxial Cross-Check

A GAA gate is not a parallel-plate capacitor. For a coaxial stack the capacitance per unit length is [1]

$$C' = \frac{2\pi\epsilon_0}{\sum_i \frac{1}{k_i}\ln\frac{r_i}{r_{i-1}}}, \tag{4}$$

so each increment of dielectric at radius r contributes $dr/(k\,r)$ to $1/C'$, rather than dr/k. Matching C' to an SiO_2 *shell* of thickness $\mathrm{EOT}_{\mathrm{coax}}$ grown on the same wire,

$$\frac{1}{k_{ox}}\ln\frac{R_1 + \mathrm{EOT}_{\mathrm{coax}}}{R_1} = \frac{1}{k_{IL}}\ln\frac{R_2}{R_1} + \frac{1}{k_{HK}}\ln\frac{R_3}{R_2}, \tag{5}$$

which inverts to

$$\mathrm{EOT}_{\mathrm{coax}} = R_1\left(e^{k_{ox}\Lambda} - 1\right), \quad \Lambda = \frac{1}{k_{IL}}\ln\frac{R_2}{R_1} + \frac{1}{k_{HK}}\ln\frac{R_3}{R_2}. \tag{6}$$

For the stack above, $\Lambda = 0.062142$ and $\mathrm{EOT}_{\mathrm{coax}} = 0.8227$ nm, i.e. **8.6% below** the planar figure. The HfO_2 occupies the outer radii, where the field lines have already diverged, so each of its nanometres is capacitively cheaper than the planar formula assumes; the physical stack therefore has slightly *more* gate capacitance than EOT $= 0.9$ nm implies. The two definitions converge as $R_1 \to \infty$: repeating the calculation at $R_1 = 100$ nm gives $\mathrm{EOT}_{\mathrm{coax}} = 0.896$ nm, within 0.4%.

I have reported the *planar* EOT of 0.900 nm, since that is the industry-conventional definition used when specifying gate stacks, and the one against which the 0.9 nm constraint is naturally read. Had the coaxial definition been intended, $t_{HK} = 2.944$ nm would be required.

C. Choice of Interfacial Layer Thickness

$t_{IL} = 0.5$ nm is deliberately chosen. It sits above the $\sim$0.4 nm physical limit of a continuous SiO_2 film, which is reachable only through remote scavenging of the interfacial oxide [2], and it is comparable to the interfacial layers of fabricated GAA nanowire devices. Thinning the IL at fixed EOT has two opposing effects that are both active in this simulation: the channel moves closer to the HfO_2, strengthening remote phonon scattering and lowering mobility, while more of the stack shifts to large radius, raising the true coaxial gate capacitance. For a 15 nm gate the electrostatic gain would likely win, but the model omits the gate leakage and interface-trap penalties that set the practical floor, so the baseline is kept at a fabricable value.

III. Doping Engineering

The channel is unintentionally doped p-type at 10^{15} cm^{-3}. Source/drain regions are set to $N_{SD} = 4\times10^{20}$ cm^{-3}, leaving margin below the 5×10^{20} cm^{-3} ceiling after activation. The tails under the inner spacers are analytic complementary error functions rolled in from each spacer edge,

$$N(x) = N_{SD}\,\mathrm{erfc}\!\left(\frac{|x - x_{sp}|}{L_{diff}}\right), \tag{7}$$

with $L_{diff} = 1.27$ nm. No thermal anneal is applied: a spike or laser anneal would smear the junctions several nanometres into a 15 nm gate and destroy the undoped channel. A 1 μs, 500 °C step is used solely to populate the active-dopant fields.

Fig. 2 shows the resulting profile, plotted as the magnitude of the net doping so that the p-type channel remains visible on a logarithmic axis. The peak concentration is 4.0×10^{20} cm^{-3}, below the ceiling. The metallurgical junctions, the zero crossings of net doping, fall at 14.24 nm and 28.76 nm, while the gate metal spans 14–29 nm. The n$^+$ tail therefore traverses each 4 nm inner spacer and overlaps the gate by 0.24 nm on each side. This slight overlap avoids the series-resistance

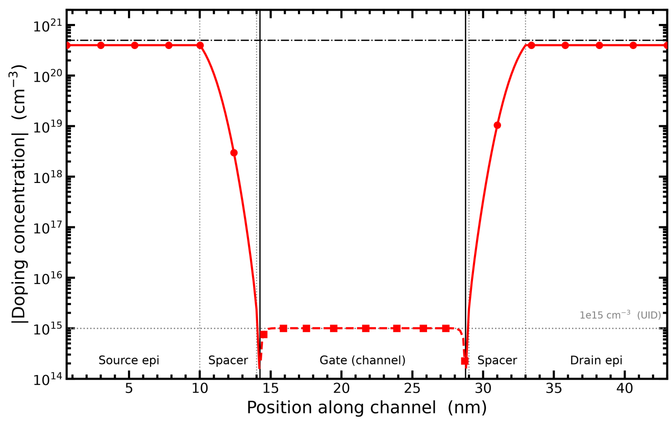

Fig. 2. Net doping magnitude along the wire axis.

penalty of an underlapped channel without incurring the gate-drain coupling and DIBL of a heavily overlapped one. The effective channel length is $L_{eff} = 14.51$ nm against a 15 nm gate metal.

IV. Physics Models and Compliance

Table II summarises the setup. Four choices warrant comment.

A. Quantum Corrections

An *isotropic* density-gradient model is applied to all silicon, including the source and drain, since a 6 nm wire is confined there too and the confinement shift sets both the injection barrier and the band-to-band tunnelling window. `AutoOrientation` is deliberately not used: it anisotropises the density-gradient equation along a fixed crystal axis, which is meaningful on a flat (100) nanosheet surface but not on a revolved cylinder whose confining normal is radial and rotates with azimuth. The requirement is that quantum corrections be enabled, not that quantisation masses be orientation-resolved.

B. Stress: Piezoresistance and the Kanda Factors

The 1 GPa uniaxial tensile stress must act on *both* the mobility and the band structure. The two are handled separately. Band-structure effects the strain-induced splitting of the conduction-band valleys, which repopulates the low-transport-mass Δ_2 valleys and shifts the density of states are captured by a deformation potential model. Stress-induced mobility change is captured by the piezoresistance tensor, whose coefficients π_{11}, π_{12}, π_{44} relate the fractional resistivity change to the applied stress.

Those coefficients, however, are tabulated for lightly doped bulk silicon. Piezoresistance weakens sharply once the semiconductor becomes degenerate, because the Fermi level moves into the band and the intervalley repopulation that drives the effect saturates. The Kanda model [3] supplies the doping and temperature-dependent factors $P_1(N,T)$ and $P_2(N,T)$ that scale the bulk coefficients; if it is not invoked, both factors are set to unity [6].

That distinction is not cosmetic in this device. The stress is applied uniformly, but the doping spans five orders of magnitude, from 10^{15} cm^{-3} in the channel to 4×10^{20} cm^{-3} in the source and drain. With $P_1 = P_2 = 1$ the simulation would apply the full lightly doped piezoresistance coefficients to the degenerate epi, and would overstate the stress-induced conductivity change in exactly the regions that set the parasitic series resistance. Enabling Kanda restores the correct doping dependence and leaves the enhancement concentrated in the undoped channel, which is where the strain is meant to act.

TABLE II
Simulation Setup and Model Choices

Aspect	Implementation
Geometry	2-D r–z region tiling, revolved by the cylindrical coordinate system
Gate stack	0.5 nm SiO_2 + 2.256 nm HfO_2, EOT = 0.900 nm (Sec. II)
Workfunction	4.4 eV, mid-gap metal gate
Spacer	Low-k ($\epsilon_r = 5$), 4 nm inner spacer
Doping	UID channel 10^{15} cm^{-3}; erfc S/D tails peaking at 4×10^{20} cm^{-3}
Stress	1 GPa tensile along transport; deformation potential for bands, Kanda piezoresistance for mobility
Transport	Drift-diffusion; $v_{sat0} = 1.07 \times 10^7$ cm/s; no ballistic correction
Mobility	Doping, normal-field, remote-phonon, thin-layer, high-field saturation
Quantisation	Isotropic density-gradient potential on all silicon
Generation	Hurkx band-to-band tunnelling [4]; gate tunnelling disabled
Carriers	Poisson + both continuity equations + both quantum potentials, fully coupled
Bias	$V_{DD} = 0.7$ V, $T = 300$ K

C. Mobility

Doping-dependent, normal-field, thin-layer and high-field saturation models are active:

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Mobility( DopingDependence Enormal(RPS)

ThinLayer(IALMob) eHighFieldSat(GradQuasiFermi) )
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The driving force for velocity saturation must be the gradient of the quasi-Fermi potential rather than the carrier temperature. The remote-phonon term captures scattering from soft optical modes in the HfO_2 [5] and is geometry-aware through the distance from the channel to the nearest high-k layer, which makes t_{IL} a genuine mobility parameter.

D. Both Carriers Are Solved

Band-to-band tunnelling generates electron hole pairs at the reverse-biased drain junction. Solving for electrons alone would leave the generated holes nowhere to go and would misreport I_{OFF}. Poisson, both continuity equations and both quantum potentials are coupled throughout.

V. Results

Fig. 3 shows the transfer characteristics at both drain biases on a logarithmic axis; Table III collects the scored metrics. The low-field sweep at $V_D = 50$ mV is used only for threshold-voltage and mobility checks, since all four competition metrics are defined at $V_D = V_{DD}$. A subthreshold swing of 63.5 mV/dec, close to the 60 mV/dec thermal limit, together with an on/off ratio of 7.54×10^5, confirms the strong

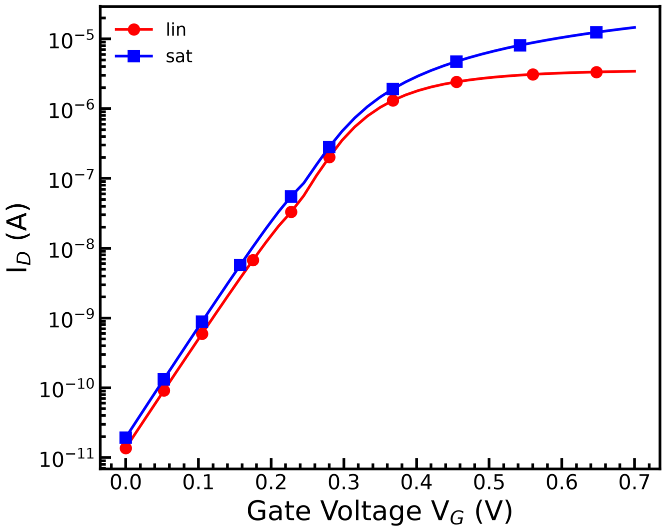

Fig. 3. I_D–V_G at $V_D = 50$ mV and $V_D = V_{DD} = 0.7$ V.

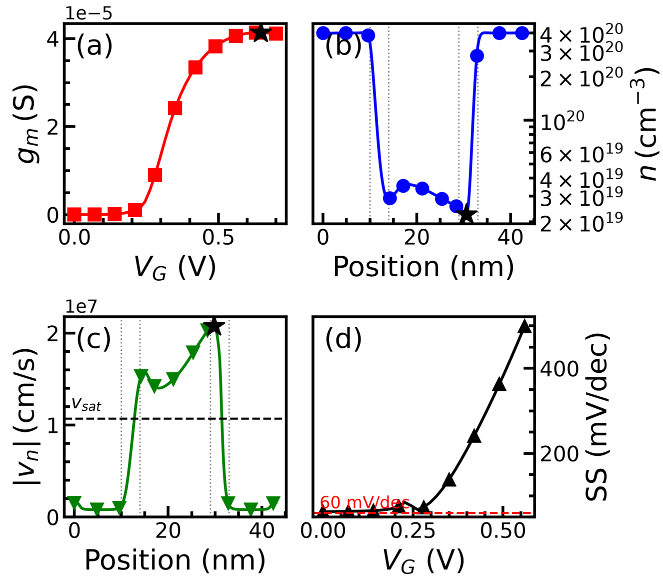

Fig. 4. Saturation-bias behaviour. (a) g_m. (b) Electron density and (c) velocity along the wire axis; stars mark the extrema, dotted lines the spacer and gate edges. (d) Subthreshold swing.

TABLE III
SCORED METRICS AT $V_D = V_G = V_{DD} = 0.7$ V

Metric	Value	Unit
(a) I_{ON}	1.459×10^{-5}	A
(b) I_{OFF}	1.936×10^{-11}	A
(c) Peak g_m	4.137×10^{-5}	S
(d) Max electron velocity	2.078×10^{7}	cm/s
I_{ON}/I_{OFF}	7.54×10^{5}	—
Subthreshold swing	63.5	mV/dec

electrostatic control expected of a gate-all-around geometry at this diameter.

A. Transport Along the Channel

Fig. 4 collects the saturation-bias behaviour. Panels (b) and (c) are taken along a single axial cutline at mid radius, $r = R/2 = 1.5$ nm, inside the inversion shell but away from both the bulk core and the semiconductor oxide interface. The dotted lines mark the inner-spacer and gate-metal edges at 10, 14, 29 and 33 nm.

The transconductance in Fig. 4(a) peaks at 4.137×10^{-5} S, below V_{DD}. That position matters for threshold extraction: linear extrapolation from the peak-g_m point yields $V_T = 0.347$ V, whereas the constant-current criterion at $I_{ref} = 1.257 \times 10^{-7}$ A gives $V_T = 0.256$ V. The two disagree by 91 mV because the drain current has already begun to flatten under velocity saturation by the time g_m peaks, biasing the extrapolated intercept high. The constant-current value is quoted.

The electron density in Fig. 4(b) drops by more than an order of magnitude on entering the channel, from 4×10^{20} cm^{-3} in the source epi to a minimum of 2.2×10^{19} cm^{-3} near the drain. That minimum sits beside the velocity peak in Fig. 4(c), and the pairing is analysed in Sec. V-B.

Fig. 4(d) plots the subthreshold swing, the reciprocal slope of the logarithmic transfer curve, whose minimum over the subthreshold window is 63.5 mV/dec. This lies within 6% of the 60 mV/dec room temperature thermal limit, shown dashed. For a drift-diffusion device with no subthermal injection mechanism, that is close to the best attainable, and it is the clearest single indicator that the gate-all-around geometry is working at $D = 6$ nm: every point of the cross-section lies within 3 nm of the gate, so the body is fully depleted and the body factor is near unity. The rise in SS above threshold is not physical degradation; it simply reflects that the swing is defined only in the subthreshold regime.

B. On the Reported Electron Velocity

The peak electron velocity, 2.078×10^7 cm/s, is 1.94 times the saturation velocity $v_{sat} = 1.07 \times 10^7$ cm/s asserted in the parameter file. This is not a modelling error, and the mechanism is visible in Fig. 4 itself.

The simulator defines the plotted quantity as the ratio of electron current density to electron charge density [6],

$$v_n = \frac{J_n}{n\,q}, \qquad J_n = q\,n\,\mu_n \mathbf{E} + q\,D_n \nabla n. \tag{8}$$

Only the first term of J_n is a drift current, and only that term is bounded by v_{sat} through the high-field mobility model. The second is diffusive. Wherever the carrier density collapses while the current density stays continuous the definition of pinch-off the quotient $J_n/(nq)$ grows without any drift velocity exceeding its cap.

The measured profiles confirm this directly. The velocity peak in Fig. 4(c) falls at $x = 29.8$ nm and the electron-density minimum in Fig. 4(b) at $x = 30.6$ nm: the extrema lie 0.8 nm apart, both just beyond the drain-side gate edge at 29 nm. The reported velocity exceeds v_{sat} over $12.8 \le x \le 31.4$ nm.

The anticorrelation is quantitative. Between the source junction at $x = 10$ nm and the density minimum, the electron density falls by a factor of 15.4, from 3.45×10^{20} to 2.24×10^{19} cm^{-3}, while the velocity rises by a factor of

TABLE IV
LINEAR AND SATURATION CHARACTERISTICS

Quantity	$V_D = 50$ mV	$V_D = 0.7$ V	Unit
I_{ON}	3.436×10^{-6}	1.459×10^{-5}	A
I_{OFF}	1.370×10^{-11}	1.936×10^{-11}	A
I_{ON}/I_{OFF}	2.51×10^{5}	7.54×10^{5}	—
Peak g_m	1.512×10^{-5}	4.137×10^{-5}	S
SS	63.52	63.49	mV/dec
V_T (const. curr.)	0.2663	0.2560	V
V_T (extrap.)	0.2808	0.3471	V
DIBL = 15.9 mV/V			

16.2, from 1.18×10^6 to 1.92×10^7 cm/s. Their product nv, proportional to the current density, changes by only 6% across the entire channel. This is current continuity: J_n is fixed by the terminal current, so $v_n = J_n/(nq)$ must rise wherever n falls, irrespective of any bound on the drift component.

Two consequences follow. The reported figure is the correct answer to the metric as posed the maximum electron velocity in the device, but it is not a drift velocity, and it does not indicate that velocity saturation is inactive or mis-calibrated. And the drift-diffusion formulation remains self-consistent with constraint 13: $v_{sat0} = 1.07 \times 10^7$ cm/s is asserted explicitly and the ballistic mobility model is absent from the parameter file. Had a ballistic correction been active, the drift component itself would have exceeded v_{sat} and the on-current would have been inflated accordingly.

The velocity is reported as the transport (y) component of the velocity vector. The radial (x) component is of order 10^4 cm/s throughout, three orders of magnitude smaller, as expected in a cylindrically symmetric device at steady state, where no net radial current flows.

C. Electrostatic Integrity

Table IV compares the two drain biases. The constant-current threshold shifts by only 10.3 mV as V_D rises from 50 mV to 0.7 V, giving a drain-induced barrier lowering of

$$\mathrm{DIBL} = \frac{V_{T,\mathrm{lin}} - V_{T,\mathrm{sat}}}{V_{DSAT} - V_{DLIN}} = 15.9 \text{ mV/V}. \qquad (9)$$

For a 15 nm gate this is small, and it is the quantitative counterpart of the near-thermal swing: the gate encircles the channel completely, so the drain field terminates on the gate rather than on the source barrier. The subthreshold swing itself is essentially bias-independent, changing by 0.03 mV/dec between the two conditions, which is a further sign that short-channel effects are not encroaching.

The table also settles the threshold-extraction question raised earlier. The two V_T definitions differ by only 15 mV at $V_D = 50$ mV, but by 91 mV at $V_D = V_{DD}$. The linear-extrapolation method assumes the above-threshold current is linear in V_G; under saturation, velocity saturation flattens it well before V_{DD}, so the extrapolated intercept is pushed high. The constant-current value is therefore the one quoted throughout.

Normalising the peak transconductance by the channel circumference, $W_{eff} = \pi D = 18.85$ nm, gives 2.195 mS/μm. This is close to the 2.15 mS/μm reported for a fabricated GAA silicon nanowire device with a comparable unscavenged gate stack, though the comparison should be read with care: normalising a cylindrical device by its circumference is a convention, not a measurement, and a footprint-pitch normalisation would give a different figure.

D. Model Scope

Three omissions bound the interpretation of these results. Gate leakage is disabled, so the thin interfacial layer carries no tunnelling penalty; interface traps and remote Coulomb scattering from fixed charge at the IL/high-k interface are absent, so the mobility is optimistic; and the reported maximum velocity is sampled along axial cutlines rather than over every mesh vertex. Each of these would, if included, act to reduce the figures of merit reported here rather than to increase them.

VI. CONCLUSION

A 2-D cylindrical GAA silicon nanowire nFET meeting all fourteen constraints has been simulated. The gate stack achieves a conventional EOT of 0.900 nm from a 0.5 nm interfacial layer and 2.256 nm of HfO_2. Spacer doping engineering places the junctions 0.24 nm inside the gate edges, yielding $L_{eff} = 14.51$ nm. The device delivers $I_{ON} = 1.459 \times 10^{-5}$ A at $I_{OFF} = 1.936 \times 10^{-11}$ A, a peak transconductance of 4.137×10^{-5} S, and a peak electron velocity of 2.078×10^7 cm/s, with a subthreshold swing of 63.5 mV/dec and DIBL of 15.9 mV/V.

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